

Flipped Classroom Exercise – Module 7: NoSQL Database Management

Module:7	NOSQL Database Management	3 hours
Introduction, Need of NoSQL, CAP Theorem, different NoSQL data bases: Key-value data stores, Columnar families, Document databases, Graph databases		

Instructions to Students:

You are required to explore the concepts using textbooks, research papers, or credible online resources and write detailed answers.

- “Answers should be written in a structured manner with diagrams wherever applicable.”
- “Real-world examples will carry additional weightage.”

Q1. Need for NoSQL & Comparison (Concept + Analysis)

- Explain the limitations of traditional relational databases that led to the emergence of NoSQL databases.
- Compare SQL and NoSQL databases based on:
 - Schema design
 - Scalability
 - Data consistency
 - Use cases

Q2. CAP Theorem (Concept + Application)

- Explain the **CAP Theorem** with a suitable diagram.
- Discuss the trade-offs between:
 - Consistency
 - Availability
 - Partition Tolerance
- Give one real-world example for each of the following systems:
 - CP system
 - AP system

Q3. Key-Value & Column Family Databases (Understanding + Differentiation)

- a) Explain the working of **Key-Value stores** with an example application.
- b) Describe **Column Family databases** and how they differ from relational tables.
- c) Compare both models based on structure, query capability, and performance.

Q4. Document Databases (Application-Oriented)

- Explain the structure and working of **Document-oriented databases**.
 - Design a sample document schema (JSON format) for an **E-commerce Order System** including:
 - Customer details
 - Order items
 - Payment details
 - Discuss advantages of document databases in such applications.
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Q5. Graph Databases (HOTS – Thinking Type)

- Consider a **social networking system** where users are connected through friendships, likes, and follows.
- Explain how this system can be modeled using a **Graph Database** (nodes, edges, properties).
 - Compare graph databases with relational databases for this use case.
 - Justify why graph databases are preferred for highly connected data.
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Overall Questions

- Define NoSQL databases. State any two reasons for their emergence.
- State the three components of the CAP Theorem.
- What is a Key-Value store? Give one example use case.
- Differentiate between Document databases and Column Family databases (any two points).
- What are nodes and edges in a Graph database?
- Case Study:**

A company “QuickCart” is building a large-scale online platform similar to Amazon. The system must handle:

- Millions of users browsing products simultaneously
- Flexible product attributes (different categories have different fields)
- Real-time recommendations based on user interactions
- Social features like reviews, likes, and user connections

Questions:

- Identify and justify the most suitable NoSQL database types (minimum TWO) for this system. (Example: Document, Graph, Key-Value, Column Family)
- Design a high-level data model for ONE chosen database type. (Example: JSON for document DB OR node-edge structure for graph DB)
- Analyze how CAP theorem trade-offs would influence the design of this system.